

# Supplementary Materials for TCRBuilder: Multi-state T-cell receptor structure prediction

## Supplementary Method

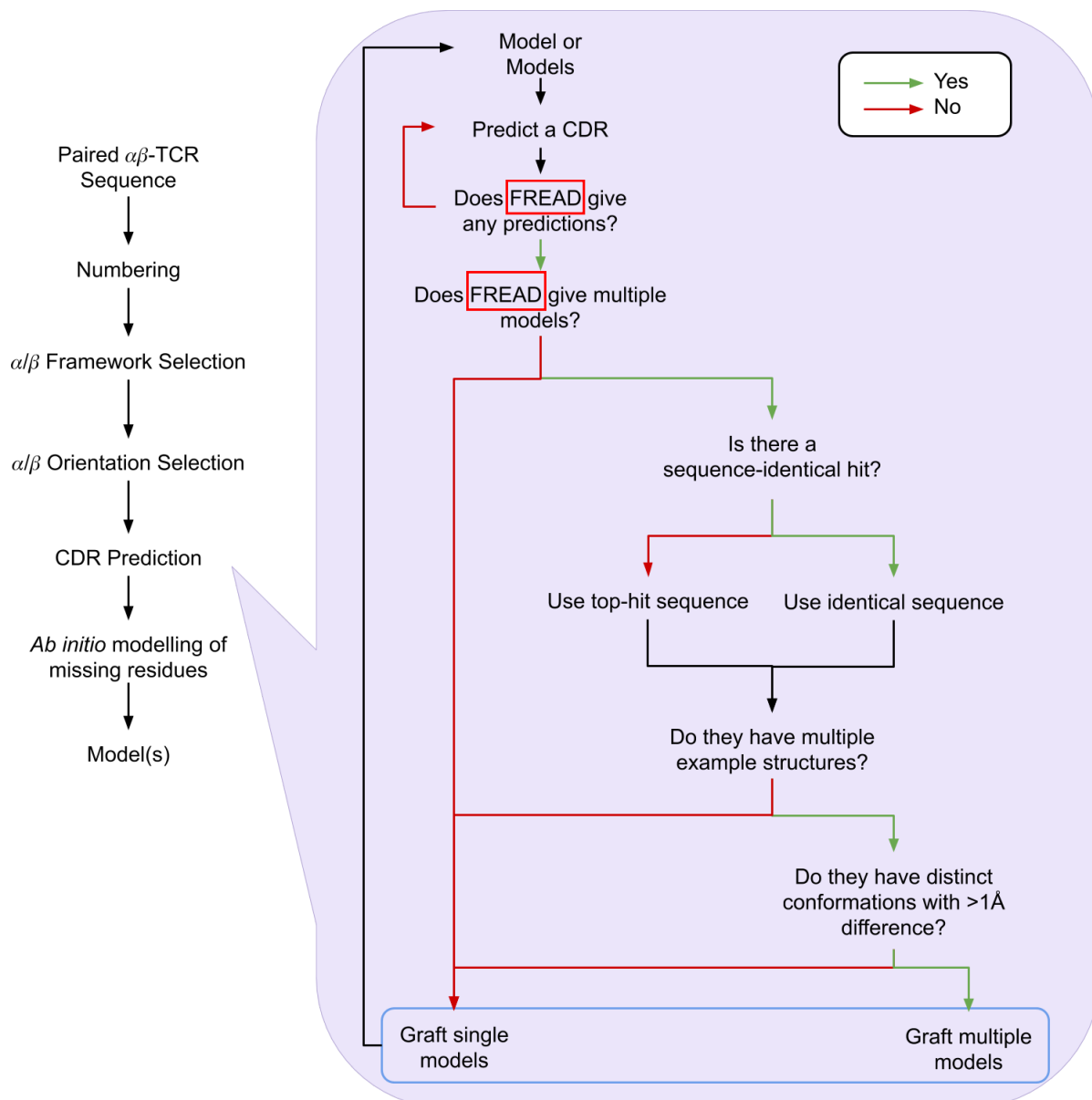
### Database

A database of  $\alpha\beta$  T-cell receptor (TCR) structures that were solved by X-ray crystallography, at resolution  $\leq 3.2$  Å, were extracted from STCRDab (Leem *et al.*, 2017), on 23 July 2019.

When building the CDR type-specific FREAD databases (Deane and Blundell, 2001; Choi and Deane, 2010), IMGT-CDR structures (Lefranc *et al.*, 2014) were extracted with five residues before and five after the CDR. Fragment lengths range between 3 and 30.

### Method

The modelling pipeline of TCRBuilder is based on the ABodyBuilder pipeline (Figure S1; Leem *et al.*, 2016).



**Figure S1. TCRBuilder pipeline for modelling TCR from sequence.** The target  $\alpha\beta$  TCR sequences are numbered by ANARCI (Dunbar and Deane, 2016). Framework and orientation templates are selected by sequence identity. Where possible, CDR templates are predicted by FREAD (Deane and Blundell, 2001; Choi and Deane, 2010) and multiple conformations of the hits are grafted as appropriate.

### Framework Selection

The  $\alpha\beta$  TCR target sequences are numbered using ANARCI (Dunbar and Deane, 2016). For each individual query chain, a framework template is selected from STCRDab (Leem *et al.*, 2017) using the highest sequence identity over the framework region. The orientation template is then chosen from the structure with the highest full sequence identity (across both the framework and the CDRs) over both chains in the pair. Any missing residues in the framework are modelled using MODELLER (Webb and Sali, 2014).

## CDR Selection

CDRs are modelled in the following order:  $\alpha 1$ ,  $\beta 1$ ,  $\alpha 2$ ,  $\beta 2$ ,  $\alpha 3$  and finally  $\beta 3$ . FREAD (Deane and Blundell, 2001; Choi and Deane, 2010), a database-based loop prediction method, searches through its database of the corresponding CDR type and returns a list of hits. If the exact query CDR sequence, or the sequence of the top hit (ranked by the anchor root-mean-square-distance; RMSD) is found in multiple distinct conformations, all unique conformations are retained and considered as alternative models. That is, if two structurally distinct templates were found for  $\alpha 1$ , each of the structures will proceed through an independent modelling pipeline of the  $\beta 1$ ,  $\alpha 2$  and so on. If there is only one conformation with an exact sequence match, or the top hit if there is no exact match, one model is selected to proceed with the modelling. Here distinct conformations refer to unique conformations whose backbone atom (C, C $\alpha$ , N and O) RMSDs differ by >1 Å.

Where FREAD makes no prediction, *ab initio* modelling of CDRs using Sphinx (Marks *et al.*, 2017) is used.

Side-chains are modelled by PEARS (Leem *et al.*, 2018), using a TCR-specific database. We used the same STCRDab set extracted on the 23 July 2019. Structures were clustered at 90% sequence identity. This resulted in a set of 123 structures, from which the database was constructed. See Leem *et al.* (2018) for the full methodology.

When Sphinx is used, for performance reasons, we only remodel the side-chains of the loop grafted on by Sphinx, while keeping the rest of the side-chains unchanged.

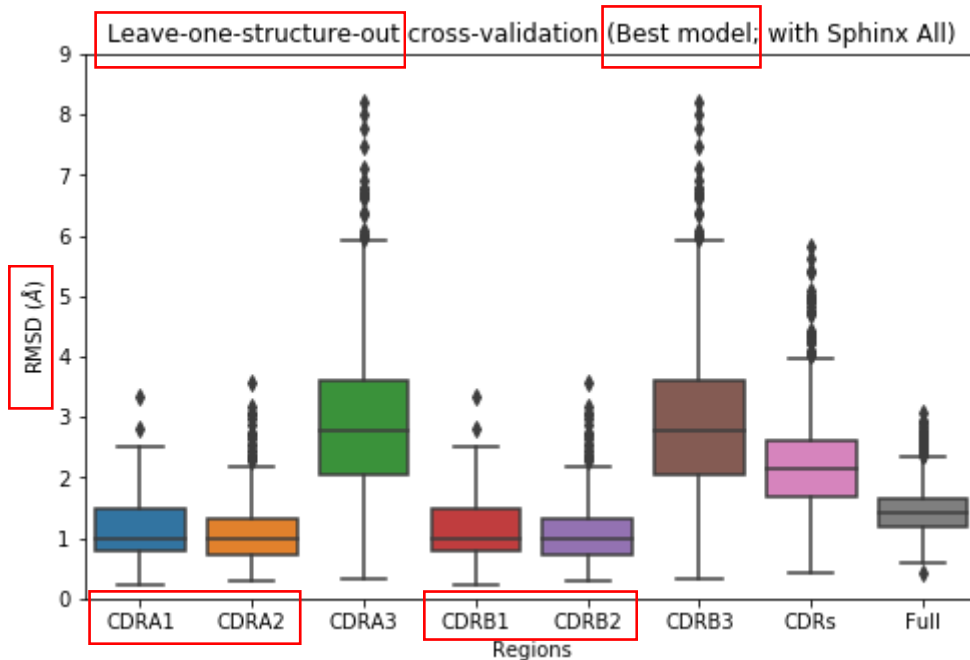
## Supplementary Results

### Cross-validation performance

TCRBuilder may produce more than one model if the given CDR sequences are known to adopt more than one conformation. We compared the performance of the best model out of all the predicted models, to the native structure. While only one conformation was observed in a native structure, the predicted models might suggest CDR conformations that were previously unseen due to the low number of available TCR structures.

We evaluated all predicted models against all native structures with the same sequence, and select the best model for each structure. The backbone RMSD was calculated across the region of interest (Figure S2). For CDRs, the backbone RMSD was calculated after aligning the two residues immediately before, and two after, the given CDR loop.

Figure S2 shows that CDR1 and 2 on both chains can be modelled with reasonable accuracy, while the accuracy for CDR3 is lower. This not only illustrates the high structural variability in CDR3, but also highlights the possibility that the predicted structure may be a correct conformation, but has not been observed due to a lack of experimental data.



**Figure S2.** The root-mean-square-distance (RMSD) of all backbone atoms across the respective regions, between the native structure and the model with the lowest RMSD.

#### Blind set performance

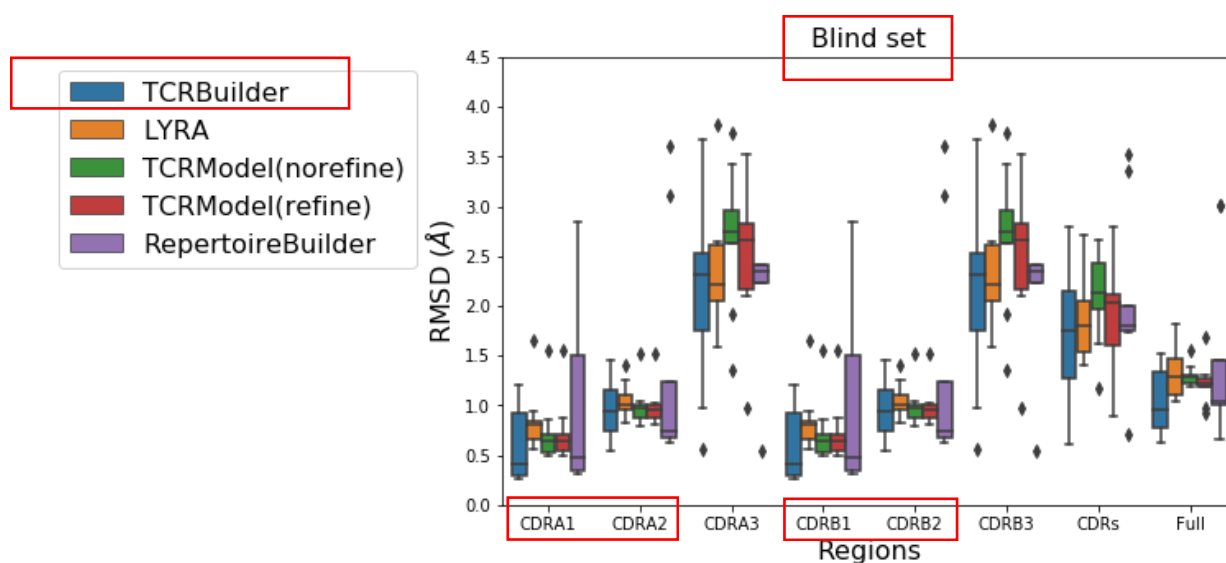
To test if **TCRBuilder** is comparable to current **TCR** modelling tools, we constructed a blind test set of the 10 (9 non-redundant)  $\alpha\beta$ TCRs deposited in the PDB between 24 July 2019 - 22 October 2019. We tested LYRA and Repertoire **Builder** using their normal modes.

**TCRModel** provides two modes for users. The results with no refinement are called “(norefine)”, and the ones with refinement are referred to as “(refine)”.

**Table S1** covers the PDB codes tested in this blind set. Figure S3 shows that **TCRBuilder** has comparable performance to the other four modelling web servers.

**Table S1.** List of PDB structures used in the blind set, and any structures that share identical sequences where applicable. PDB codes are listed as PDB ID followed by the TCR chain IDs.

| PDB Codes in the blind set (and structures with identical sequences) |
|----------------------------------------------------------------------|
| 6MSS_BA (6MRA_BA)                                                    |
| 6MNM_BA                                                              |
| 6MKR_BA                                                              |
| 6MKD_BA                                                              |
| 6JXR_nm                                                              |
| 6MNO_BA                                                              |
| 6MNN_BA                                                              |
| 6MNG_BA                                                              |
| 6Q3S_ED                                                              |



**Figure S3.** Benchmark performance on the blind set with other existing TCR modelling web servers.

## References

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